NAME: _____ Geometry (FL B.E.S.T.)

8.1 Distance, Midpoint, and Slope

Three formulas describe a segment completely on the coordinate plane. The **distance formula**, $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$, is just the Pythagorean Theorem applied to horizontal and vertical legs. The **midpoint formula**, $M = (\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2})$, averages the coordinates. The **slope formula**, $m = \frac{y_2 - y_1}{x_2 - x_1}$, measures steepness and direction, and lets you test whether two segments are parallel (equal slopes) or perpendicular (slopes multiply to -1). A horizontal segment has slope 0; a vertical segment has undefined slope. This worksheet also uses the midpoint formula in reverse to find a missing endpoint, and uses slope to test whether three points lie on the same line.

Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

1. Distance between $(0, 0)$ and $(6, 8)$.

2. Midpoint of $(2, 9)$ and $(-4, 3)$.

3. Slope through $(-2, 4)$ and $(3, -6)$.

4. Midpoint $(5, -2)$, $P = (9, 1)$. Find Q.

5. Slopes 2 and 2. Parallel?

6. Slopes $\frac{2}{5}$ and $-\frac{5}{2}$. Perpendicular?

7. Between $(1, 1)$ and $(4, 5)$.

8. Of $(0, 0)$ and $(8, -6)$.

9. Through $(3, 7)$ and $(3, -2)$.

10. Through $(-1, 4)$ and $(5, 4)$.

11. Midpoint $(0, 0)$, $A = (4, -2)$. Find B.

NAME: _____ Geometry (FL B.E.S.T.)

12. Between $(-3, 2)$ and $(1, 5)$.

13. Slopes 1 and 1. Perpendicular?

14. $\overline{AB}$ has $A(-3, 4)$ and midpoint $M(1, 1)$. Find endpoint B, then find the length AB.

15. Show that $A(0, 0)$, $B(2, 4)$, $C(5, 10)$ are collinear, then verify $AB + BC = AC$.

✓ Answer Key

1. 10
2. $(-1, 6)$
3. -2
4. $(1, -5)$
5. Yes
6. Yes
7. 5
8. $(4, -3)$
9. Undefined
10. 0
11. $(-4, 2)$
12. 5
13. No
14. $B = (5, -2)$; $AB = 10$
15. Collinear (all slopes $= 2$); $AB = 2\sqrt{5}$, $BC = 3\sqrt{5}$, $AC = 5\sqrt{5}$, and $2\sqrt{5} + 3\sqrt{5} = 5\sqrt{5}$